

Three LES cases

BOMEX (ocean), SCMS and ARM (land).

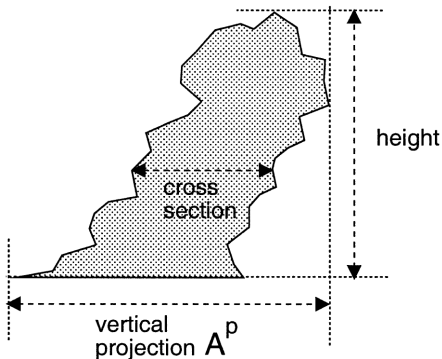
Cloud: connected saturated cells.

$$l = \sqrt{A^p}$$

$$N = \int_0^\infty \mathcal{N}(l) dl$$

$\mathcal{N}(l)$: cloud count per unit size.

A^p : vertically projected cloud area.



Cloud object



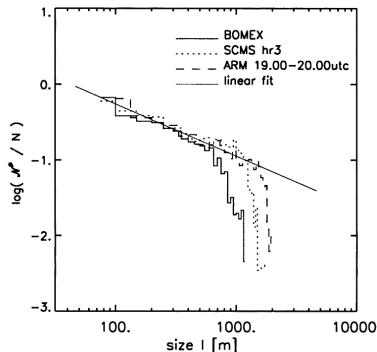
Projection A^p



Size histogram

A power law below a variable size break

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$$\mathcal{N}^* = l \ln(10) \mathcal{N} \quad (\text{log bins})$$

Size density below ℓ_d

$$\mathcal{N}(l) \propto l^{-1.70}$$

About one decade.
A steeper fall above ℓ_d .

Break ℓ_d

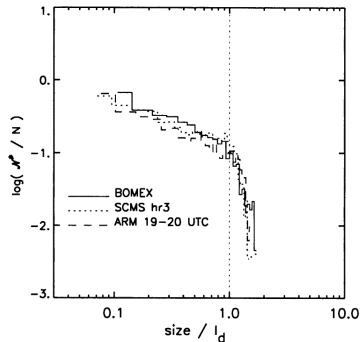
BOMEX	700 m
SCMS	1050 m
ARM	400–1250 m

The break varies with case and time of day

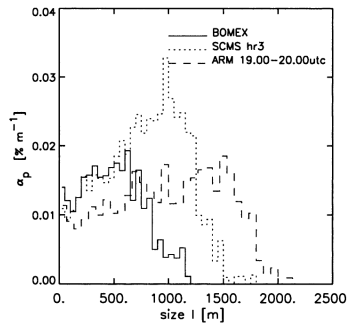
The size break sets the dominant cloud cover

3

Rescaling by ℓ_d : collapse



$$\alpha^p(l) = \frac{l^2 \mathcal{N}(l)}{S}$$

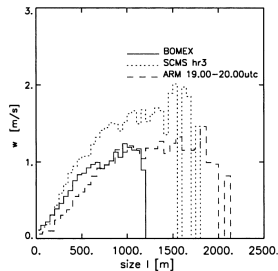


Area weighting gives $\alpha^p \propto l^{b+2}$: the cover peaks near ℓ_d

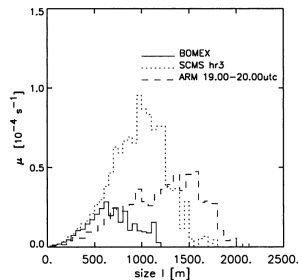
Larger cloud velocities shift the transport peak

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Mean vertical velocity



Mass-flux contribution



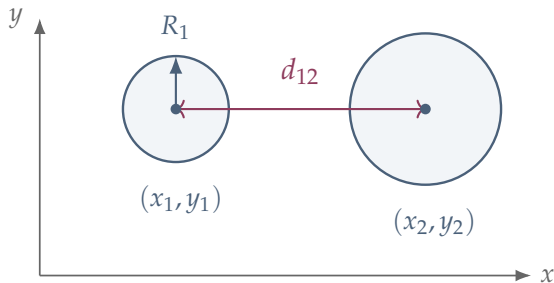
Small clouds carry little flux.

$$\mu(l) = \overline{\alpha(l, z) w(l, z)}$$

Mass flux peaks at larger cloud sizes than projected cover

Reported: projected cloud sizes, $\mathcal{N}(l)$.

Spatial quantities (schematic)



Not reported for thermals

Size distribution R_i

Positions (x_i, y_i)

Distances d_{ij}

Projected cumulus size does not determine thermal size or spacing